

CAB222 - Assessment 2

Group 59:

Stalin Baby (12254959)
Ivan Ostapenko (11421860)
Ethan Ember (11959142)
Don Philip Binu (12083623)

Normal TCP Behaviour

TCP facilitates communication through three main stages:

1. Connection establishment

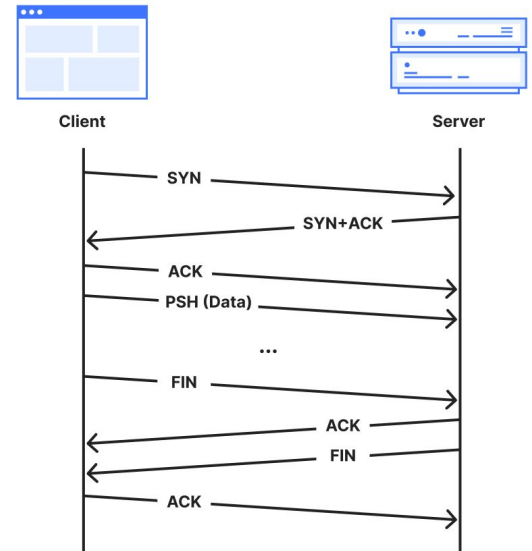
Connection is established in the form of a three-way handshake.

2. Data Transfer

Data is then broken up into smaller segments and transmitted over the network.

3. Connection Termination

The connection is closed in the form of a four-way handshake.



Normal TCP Behaviour

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.40.40	192.168.30.30	TCP	74	33946 → 80
2	0.000018	192.168.30.30	192.168.40.40	TCP	74	80 → 33946
3	0.001004	192.168.40.40	192.168.30.30	TCP	66	33946 → 80

Connection Establishment: This sequence is a three-way handshake between Gail (192.168.40.40) and the HTTP server (192.168.30.30).

4	0.001004	192.168.40.40	192.168.30.30	HTTP	203	GET /power.csv HTTP/1.1
5	0.001017	192.168.30.30	192.168.40.40	TCP	66	80 → 33946 [ACK] Seq=1 Ack=:
6	0.001359	192.168.30.30	192.168.40.40	HTTP	2504	HTTP/1.1 200 OK (text/csv)

Data transfer: GET /power.csv HTTP/1.1" confirms application-layer data transfer. The server then acknowledges the request and sends data.

7	0.001703	192.168.40.40	192.168.30.30	TCP	66	33946 → 80 [ACK] Seq=1
8	0.002211	192.168.40.40	192.168.30.30	TCP	66	33946 → 80 [FIN, ACK]
9	0.002392	192.168.30.30	192.168.40.40	TCP	66	80 → 33946 [FIN, ACK]
10	0.002450	192.168.40.40	192.168.30.30	TCP	66	33946 → 80 [ACK] Seq=1

Connection Termination: Sessions end with correct FIN/ACK exchanges (four-way handshake), showing that TCP is functioning correctly.

Traffic Abnormalities and Potential Attacks

No.	Time	Source	Destination	Protocol	Length	Info
30	0.449407	192.168.30.10	192.168.30.30	TCP	66	58072 → 80 [ACK]
31	0.456234	192.168.30.10	192.168.30.30	TCP	1078	39762 → 80 [FIN]
32	0.500524	192.168.30.10	192.168.30.30	TCP	1078	58665 → 80 [FIN]
33	0.528442	192.168.30.10	192.168.30.30	TCP	1078	60754 → 80 [FIN]
34	0.560910	192.168.30.10	192.168.30.30	TCP	1078	2255 → 80 [FIN]
35	0.592606	192.168.30.10	192.168.30.30	TCP	1078	24471 → 80 [FIN]
36	0.624586	192.168.30.10	192.168.30.30	TCP	1078	44186 → 80 [FIN]
37	0.656643	192.168.30.10	192.168.30.30	TCP	1078	47049 → 80 [FIN]
38	0.688850	192.168.30.10	192.168.30.30	TCP	1078	14941 → 80 [FIN]
39	0.721448	192.168.30.10	192.168.30.30	TCP	1078	60776 → 80 [FIN]
40	0.753735	192.168.30.10	192.168.30.30	TCP	1078	5824 → 80 [FIN]
41	0.785536	192.168.30.10	192.168.30.30	TCP	1078	23699 → 80 [FIN]
42	0.816426	192.168.30.10	192.168.30.30	TCP	1078	43888 → 80 [FIN]

- **Out of Place FIN Packets**

- Packets with FIN flag sent at small intervals from same IP address
- The packets are not part of 4-way handshake indicating connection closure
- Abnormal frame data, 1024 bytes, containing 1024 repeating ASCII symbols

0030	20 00 f3 74 00 00 41 41	41 41 41 41 41 41 41 41	..t..AA	AAAAAAAA
0040	41 41 41 41 41 41 41 41	41 41 41 41 41 41 41 41	AAAAAAAA	AAAAAAAA
0050	41 41 41 41 41 41 41 41	41 41 41 41 41 41 41 41	AAAAAAAA	AAAAAAAA
0060	41 41 41 41 41 41 41 41	41 41 41 41 41 41 41 41	AAAAAAAA	AAAAAAAA
0070	41 41 41 41 41 41 41 41	41 41 41 41 41 41 41 41	AAAAAAAA	AAAAAAAA
0080	41 41 41 41 41 41 41 41	41 41 41 41 41 41 41 41	AAAAAAAA	AAAAAAAA
0090	41 41 41 41 41 41 41 41	41 41 41 41 41 41 41 41	AAAAAAAA	AAAAAAAA
00a0	41 41 41 41 41 41 41 41	41 41 41 41 41 41 41 41	AAAAAAAA	AAAAAAAA
00b0	41 41 41 41 41 41 41 41	41 41 41 41 41 41 41 41	AAAAAAAA	AAAAAAAA
00c0	41 41 41 41 41 41 41 41	41 41 41 41 41 41 41 41	AAAAAAAA	AAAAAAAA
00d0	41 41 41 41 41 41 41 41	41 41 41 41 41 41 41 41	AAAAAAAA	AAAAAAAA
00e0	41 41 41 41 41 41 41 41	41 41 41 41 41 41 41 41	AAAAAAAA	AAAAAAAA
00f0	41 41 41 41 41 41 41 41	41 41 41 41 41 41 41 41	AAAAAAAA	AAAAAAAA
0100	41 41 41 41 41 41 41 41	41 41 41 41 41 41 41 41	AAAAAAAA	AAAAAAAA
0110	41 41 41 41 41 41 41 41	41 41 41 41 41 41 41 41	AAAAAAAA	AAAAAAAA
0120	41 41 41 41 41 41 41 41	41 41 41 41 41 41 41 41	AAAAAAAA	AAAAAAAA
0130	41 41 41 41 41 41 41 41	41 41 41 41 41 41 41 41	AAAAAAAA	AAAAAAAA
0140	41 41 41 41 41 41 41 41	41 41 41 41 41 41 41 41	AAAAAAAA	AAAAAAAA
0150	41 41 41 41 41 41 41 41	41 41 41 41 41 41 41 41	AAAAAAAA	AAAAAAAA

- **Likely Attack: FIN Flood**

- Spoofed IP address (102.168.30.10)
- Exhausting server's resources
- DoS attack

Traffic Abnormalities and Potential Attacks

No.	Time	Source	Destination	Protocol	Length	Info
51	1.093745	192.168.40.50	192.168.30.30	TCP	1078	19962 → 80 [SYN] Seq=0
52	1.093766	192.168.30.30	192.168.40.50	TCP	58	80 → 19962 [SYN, ACK] S
53	1.094729	192.168.40.50	192.168.30.30	TCP	54	19962 → 80 [RST] Seq=1
54	1.105277	192.168.30.10	192.168.30.30	TCP	1078	14847 → 80 [FIN] Seq=1
55	1.125937	192.168.40.50	192.168.30.30	TCP	1078	47899 → 80 [SYN] Seq=0
56	1.125963	192.168.30.30	192.168.40.50	TCP	58	80 → 47899 [SYN, ACK] S
57	1.127149	192.168.40.50	192.168.30.30	TCP	54	47899 → 80 [RST] Seq=1
58	1.136378	192.168.30.10	192.168.30.30	TCP	1078	63562 → 80 [FIN] Seq=1
59	1.157739	192.168.40.50	192.168.30.30	TCP	1078	6488 → 80 [SYN] Seq=0 W
60	1.157773	192.168.30.30	192.168.40.50	TCP	58	80 → 6488 [SYN, ACK] Se
61	1.159105	192.168.40.50	192.168.30.30	TCP	54	6488 → 80 [RST] Seq=1 W
62	1.168488	192.168.30.10	192.168.30.30	TCP	1078	63707 → 80 [FIN] Seq=1
63	1.189551	192.168.40.50	192.168.30.30	TCP	1078	23668 → 80 [SYN] Seq=0

- **Dropped 3-way Handshakes**

- The connection is dropped with RST, instead of establishing with ACK
- SYN and RST packets are coming from the same IP
- Abnormal frame data of SYN packets, 1024 bytes, containing 1024 repeating ASCII symbols

- **Likely Attack: SYN Scan**

- Spoofed IP address (102.168.40.50)
- Checking available ports on server
- RST to drop connection

- **Other Possible Attacks**

- Expending server resources, DoS
- Checking if server is down

Attack Summary & Mitigation Strategy

SYN Stealth Scan

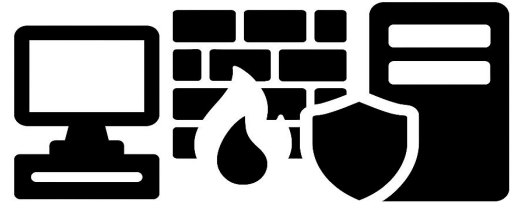
- Repeated SYN packets, some completing connections
- Goal: identify open ports or exhaust server resources

FIN Flood

- FIN packets not part of valid sessions
- Abnormal 1024-byte repeating ASCII payloads
- Goal: confuse server, disrupt normal TCP closure

Initial Defence Actions

- Firewall segmentation between workstation & server
- Intrusion Detection for abnormal SYN/FIN patterns
- Rate-limiting to prevent resource exhaustion



Network Segmentation & Access Control	Intrusion Detection & Prevention (IDPS)	Rate Limiting & SYN Scan & FIN Flood Protections
Apply strict firewall rules between workstation (192.168.40.50) and server (192.168.30.30).	Detect repeated SYNs or FINs with 1024-byte payloads.	Apply SYN cookies and per-source FIN/SYN thresholds.
Block SYN/FIN packets with abnormal size or timing.	Auto-block malicious IPs and alert operators.	Prevents DoS without major impact on real users.
Reduces lateral movement but adds configuration complexity.	Automated detection but may slow network performance.	May affect legitimate high-volume traffic.

Service Level Agreement

- The SLA formalizes the roles and responsibility of both the OT and IT department and sets clear, quantitative targets which will suffice to create transparent and accountable functioning of both departments. Thus ensuring the operations remain uninterrupted against security threats.

- Roles:

IT Department	OT Department
Continuous Network Monitoring	Report issues with minimal delay
Quick Response to Security Incidents	Provide IT with required access for monitoring
Provide Support and Timely Security Reports	Follow security recommendations by the IT department

Key Performance Indicators

KPI	Target	Justification
System Availability	99.9%	Ensure the continuous operation of the OT Systems
Critical Incident Response Time	20 Minutes	Quick containment of Security breaches
Critical Incident Resolution Time	2 Hours	Reduce system downtime
Medium/Low Incident Resolution Time	5 Hours	Resolve low priority incidents efficiently while prioritizing urgent tasks
Security Reports Delivery	Every 2 weeks	Regular System Health updates
Quarterly Vulnerability Assessments	100%	Identify vulnerabilities and prepare solutions before incidents occur
Support Availability	24/7	Rapid assistance for incidents, even outside business hours

Thank You